

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)****Y25 - 2026-2027 Trimester - 4****Project Abstract Submission Form**

Sec No - 3	Team No - 10	Project Title - Smart Waste Management System
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1. Abstract

Smart Waste Management System is a database-driven solution developed to address the inefficiencies of traditional waste collection systems that rely on fixed schedules and manual monitoring. Such practices often result in overflowing bins, unnecessary collection trips, inefficient use of manpower and fuel, and unhygienic surroundings due to the lack of timely information about waste levels. The proposed system introduces a smart, data-driven approach for monitoring waste conditions and improving collection management.

The system uses simulated real-time waste-level data to represent sensor readings and stores the data in a centralized database for monitoring and collection management. It maintains information about waste bins, their locations, capacities, waste levels, and collection status. When a bin reaches a predefined threshold, the system identifies the bin as requiring collection and supports the creation, assignment, and tracking of collection requests. Historical waste-level and collection records are also maintained to analyse waste patterns and generate useful reports.

By replacing fixed-schedule monitoring with data-driven collection management, the system aims to reduce bin overflow, minimize unnecessary collection trips, improve the utilization of manpower and fuel, and support cleaner and more sustainable communities. The project demonstrates how database technology can be integrated with smart monitoring concepts to improve the efficiency and reliability of waste management.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520030607	Ili.Harshika	
2520030365	Angela Jose	
2520030312	Chinta Pavan Kumar	

3.Frameworks / Platforms / Software Used

- MySQL – Relational database for storing and managing smart-bin, waste-level, collection, and user data.
- Node.js – Backend runtime for processing requests and simulating real-time waste-level data.
- Express.js – Backend framework for developing REST APIs.
- React.js – Frontend framework for the waste monitoring and management dashboard.
- MySQL Workbench – Database design, SQL development, and database management.
- Postman – REST API development and testing.
- Visual Studio Code – Development environment.

Guide Name and Signature (remarks if any) _____